CO1 - AT3 - CLOUD COMPUTING AND BIG DATA ANALYTICS

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Questions :-

1) Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

3) Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

4) Discuss the role of data centers in delivering cloud services.

5) Differentiate IAAS, Paas, Saas, and Xaas with one suitable real-world example for each model.

1) Define cloud computing and explain 4 characteristics.

Definition:

cloud computing is the delivery of computing resources such as servers, storage, databases, networking and software over the Internet.

characteristics:

1. On-Demand self service - users can get computing resources whenever they need them without directly contacting the service provider.
2. Broad network access - cloud services can be accessed through the Internet using laptops, mobiles, tablets etc.
3. Resource pooling - The provider shares computing resources among many users using virtualization.
4. Rapid elasticity - Resources can be quickly increased or decreased according to demand.

2) i) Mainfram era:-

Large mainframe computers were shared by many users through terminals.

ii) Personal computers : computing become available on individual computers, reducing dependance on centralized systems.

iii) Virtualization : It allow one physical server to run multiple virtual machine. This improved hardware utilization.

iv) Internet evolution : growth of internet and broadband network made it possible to access computing resources remotely.

v) Internet software evolution :- It includes web technologies, web servers, AP and service oriented Architecture.

3) Server virtualization :-

* It is a technology that divide, physical server into a multiple virtual machine.

* Each virtual machine run its own operating system and application independent.

i) Better utilization - Physical server can run multiple virtual machine.

ii) cost reduction - Reduces the need of many provider server.

iii) Scalability - Virtual machine can created, removed and resized quickly.

iv) Isolation - different user and application can run separately on same provided server.

Virtualization	cloud computing
It is a enabling technology	service delivery model
create virtual machine from physical resources	Provide computing resources through internet
Focus on resources sharing and isolation	Focus on demand service and accessibility
can exist without cloud computing	commonly use virtualization as function
Eg:- run multiple VM one one server	Eg:- using cloud server storage or software on demand

4) A data server is a facility that contains servers, storage systems, networking equipment, power systems and cooling systems

Role of Data centers:-

1) Resource hosting: Data centers contain the physical servers and storage used to provide cloud services.

2) Data storage: They store user files, databases, applications and backups securely.

3) High Availability: Multiple servers and backup systems help keep cloud services available even when hardware fails.

4) Scalability: Additional servers and storage can be added when the number of users or workload increases.

5) Networking: High speed network connections allow users to access cloud services from anywhere through the Internet.

6) cloud service:
cloud services are divided based on what cloud provide manage and what the user manage.

Iaas → Infrastructure as a service

provide virtualized computing infrastructure such as server, storage and networking

Eg:- Amazon EC2

Iaas => Get hardware resources

Paas => Platform as a service

provide a readymade platform for development to build, test and deploy application.

Paas => platform development

Eg:- google App engine.

Saas => software as a service

provide a complete software application through computing resources.